Labwork 3 - Introduction to cryptography

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Write a program in Matlab, C++ or python to solve below exercises.

Exercise 1

Given a message "flow" (in ascii code), find the corresponding integer number representing this message in Python.

Hint: how to convert a string to int in python?

Exercise 2

Given an integer number 14445 , find the corresponding message (a string) in ascii code that this integer represents in Python.

Hint: how to convert a string to int in python?

Exercise 3

Based on the provided description of RSA cryptosystem, implement RSA encryption and decryption process. System parameters p, q, e or d are given. Define keypair, ciphertext C if plaintext M is given, and reversely.

Test the program with following parameters:

1. $p = 251, q = 191, e = 65537, M = 26729$
2. $p = 233, q = 151, d = 9473, C = 29600$

Exercise 4

Find the plaintext string in ascii code, given that the ciphertext represented by an integer 412589464. The RSA parameters are set up as below: $p = 81401, q = 27109, e = 65537$